

git log/history

References

- [Visualizing Code: Polyglot Notebooks Repository \(YouTube\)](#)
- [gitstractor \(GitHub\)](#)
- [Neo4j Python Driver](#)

Command line execution (CLI mode): Yes

Git History - Directory Commit Statistics

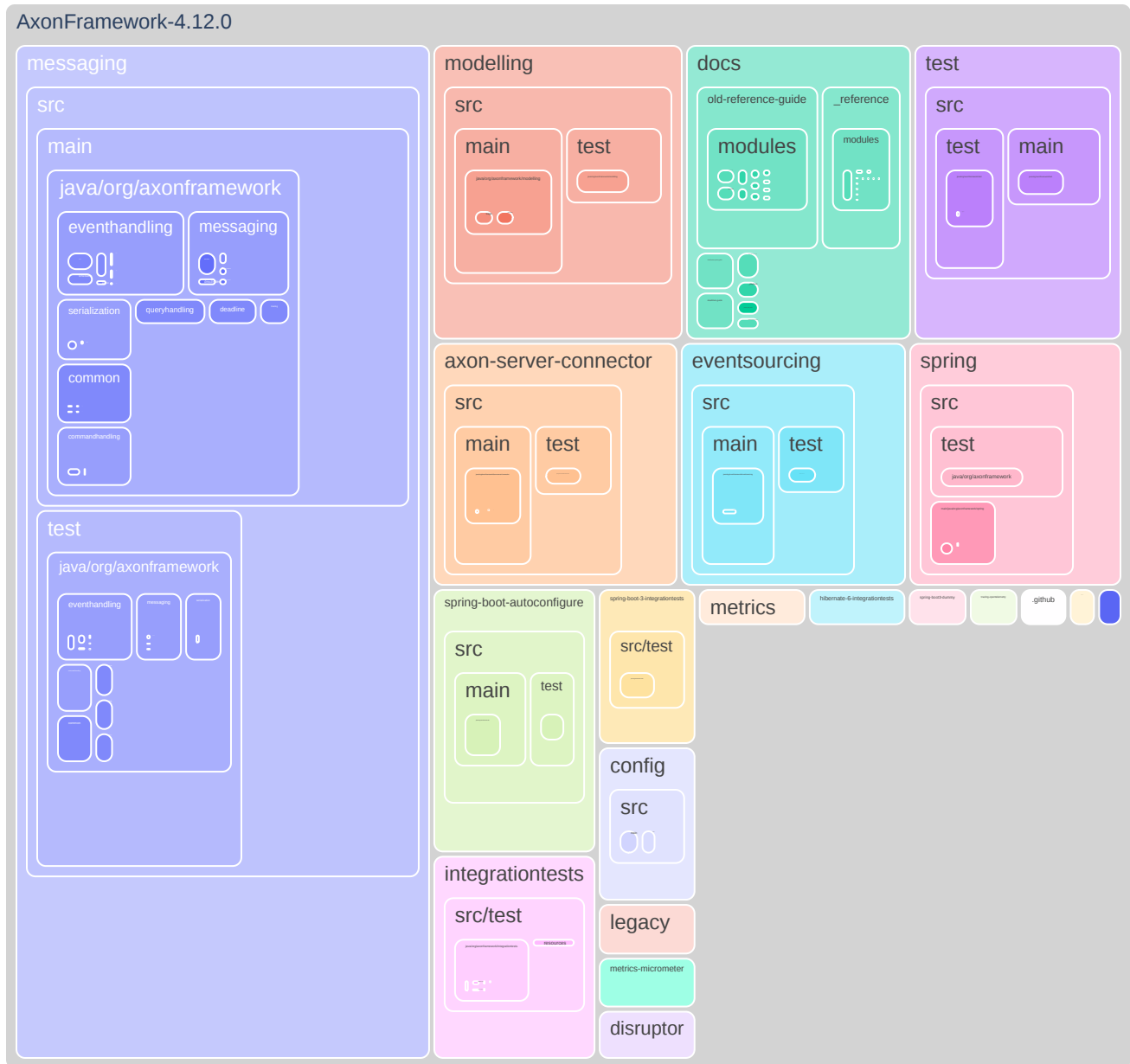
Data Preview

	fileCount	authorCount	commitCount	daysSinceLastCommit	daysSinceLastCreation	daysSinceLastModification
count	564.000000	564.000000	564.000000	564.000000	564.000000	564.000000
mean	24.898936	19.664894	288.028369	277.925532	1150.741135	371.762411
std	120.739280	21.297173	638.376287	165.621714	980.398083	389.612856
min	1.000000	2.000000	3.000000	9.000000	9.000000	8.000000
25%	1.000000	7.000000	27.000000	136.000000	410.000000	226.000000
50%	4.000000	12.000000	81.500000	369.000000	928.000000	368.000000
75%	10.000000	25.000000	270.250000	369.000000	1975.750000	368.000000
max	2294.000000	213.000000	9793.000000	2519.000000	5646.000000	2998.000000

	directoryPath	directoryParentPath
0	AxonFramework-4.12.0/assembly	AxonFramework-4.12.
1	AxonFramework-4.12.0/axon-server-connector/src/main/java/org/axonframework/axonserver/connector/heartbeat/source	AxonFramework-4.12.0/axon-server-connector/src/main/java/org/axonframework/axonserver/connector/heartbeat
2	AxonFramework-4.12.0/axon-server-connector/src/main/resources/META-INF/services	AxonFramework-4.12.0/axon-server-connector/src/main/resources
3	AxonFramework-4.12.0/axon-server-connector/src/test/java/org/axonframework/axonserver/connector/event/util	AxonFramework-4.12.0/axon-server-connector/src/test/java/org/axonframework/axonserver/connector/event
4	AxonFramework-4.12.0/config/src/test/resources/META-INF	AxonFramework-4.12.0/config/src/test/resources
5	AxonFramework-4.12.0/coverage-report	AxonFramework-4.12.
6	AxonFramework-4.12.0/docs/_playbook/localLinks	AxonFramework-4.12.0/docs/_playbook
7	AxonFramework-4.12.0/docs/_reference/modules/ROOT/attachments	AxonFramework-4.12.0/docs/_reference/modules/ROOT
8	AxonFramework-4.12.0/docs/_reference/modules/ROOT/partials	AxonFramework-4.12.0/docs/_reference/modules/ROOT
9	AxonFramework-4.12.0/docs/_reference/modules/axon-server-connector/pages	AxonFramework-4.12.0/docs/_reference/modules/axon-server-connector
10	AxonFramework-4.12.0/docs/_reference/modules/axon-server-connector/partials	AxonFramework-4.12.0/docs/_reference/modules/axon-server-connector
11	AxonFramework-4.12.0/docs/_reference/modules/configuration/pages	AxonFramework-4.12.0/docs/_reference/modules/configuration
12	AxonFramework-4.12.0/docs/_reference/modules/configuration/partials	AxonFramework-4.12.0/docs/_reference/modules/configuration
13	AxonFramework-4.12.0/docs/_reference/modules/disruptor/pages	AxonFramework-4.12.0/docs/_reference/modules/disruptor
14	AxonFramework-4.12.0/docs/_reference/modules/disruptor/partials	AxonFramework-4.12.0/docs/_reference/modules/disruptor
15	AxonFramework-4.12.0/docs/_reference/modules/eventsourcing/pages	AxonFramework-4.12.0/docs/_reference/modules/eventsourcing
16	AxonFramework-4.12.0/docs/_reference/modules/eventsourcing/partials	AxonFramework-4.12.0/docs/_reference/modules/eventsourcing
17	AxonFramework-4.12.0/docs/_reference/modules/legacy/pages	AxonFramework-4.12.0/docs/_reference/modules/legacy
18	AxonFramework-4.12.0/docs/_reference/modules/legacy/partials	AxonFramework-4.12.0/docs/_reference/modules/legacy
19	AxonFramework-4.12.0/docs/_reference/modules/metrics/pages	AxonFramework-4.12.0/docs/_reference/modules/metrics
20	AxonFramework-4.12.0/docs/_reference/modules/metrics/partials	AxonFramework-4.12.0/docs/_reference/modules/metrics
21	AxonFramework-4.12.0/docs/_reference/modules/micrometer/pages	AxonFramework-4.12.0/docs/_reference/modules/micrometer
22	AxonFramework-4.12.0/docs/_reference/modules/micrometer/partials	AxonFramework-4.12.0/docs/_reference/modules/micrometer
23	AxonFramework-4.12.0/docs/_reference/modules/modeling/partials	AxonFramework-4.12.0/docs/_reference/modules/modeling
24	AxonFramework-4.12.0/docs/_reference/modules/spring/pages	AxonFramework-4.12.0/docs/_reference/modules/spring
25	AxonFramework-4.12.0/docs/_reference/modules/spring/partials	AxonFramework-4.12.0/docs/_reference/modules/spring
26	AxonFramework-4.12.0/docs/_reference/modules/springboot-starter/pages	AxonFramework-4.12.0/docs/_reference/modules/springboot-starter
27	AxonFramework-4.12.0/docs/_reference/modules/springboot-starter/partials	AxonFramework-4.12.0/docs/_reference/modules/springboot-starter
28	AxonFramework-4.12.0/docs/_reference/modules/test/pages	AxonFramework-4.12.0/docs/_reference/modules/test
29	AxonFramework-4.12.0/docs/_reference/modules/test/partials	AxonFramework-4.12.0/docs/_reference/modules/test

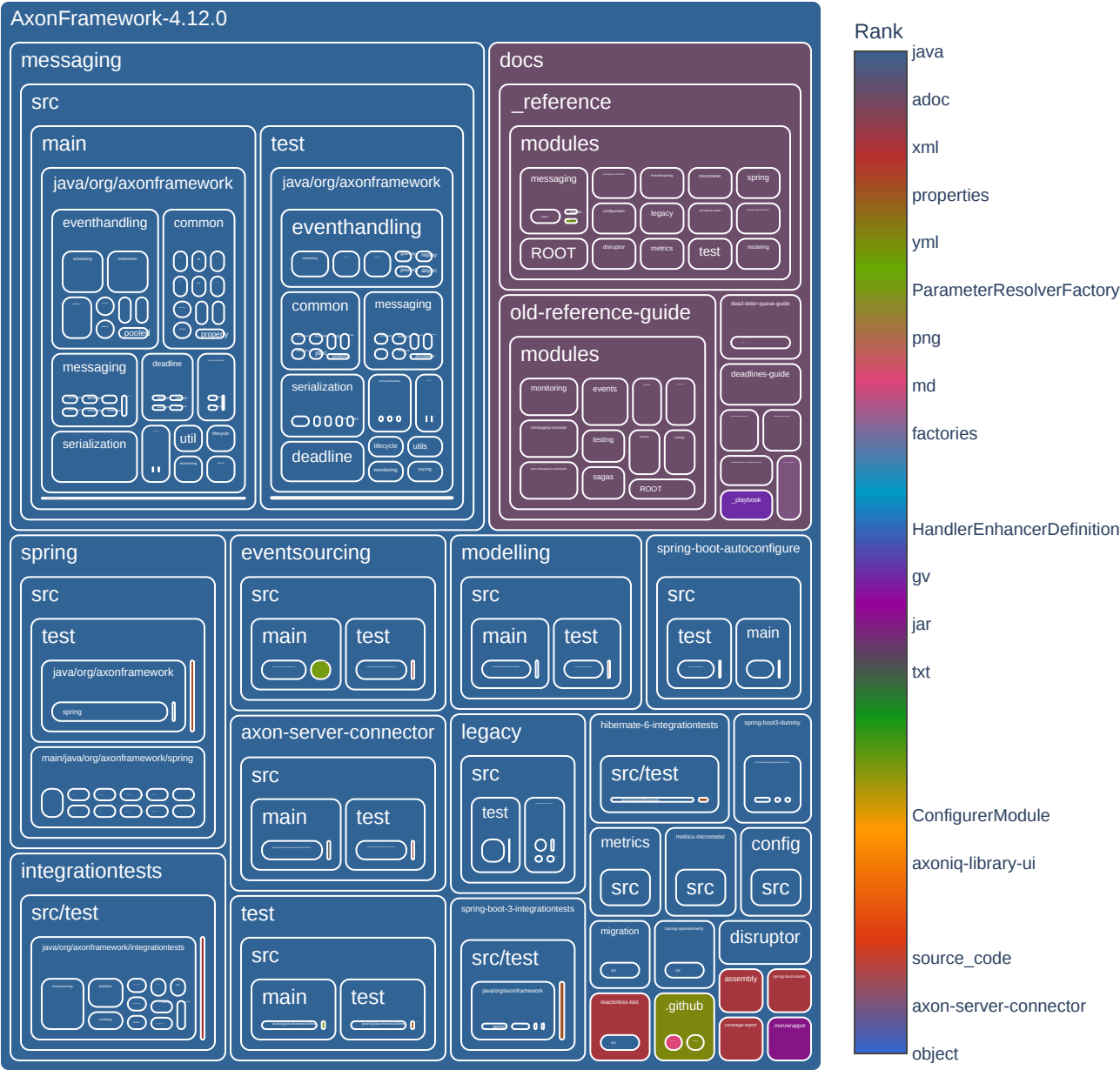
Number of files per directory

Directories and their file count



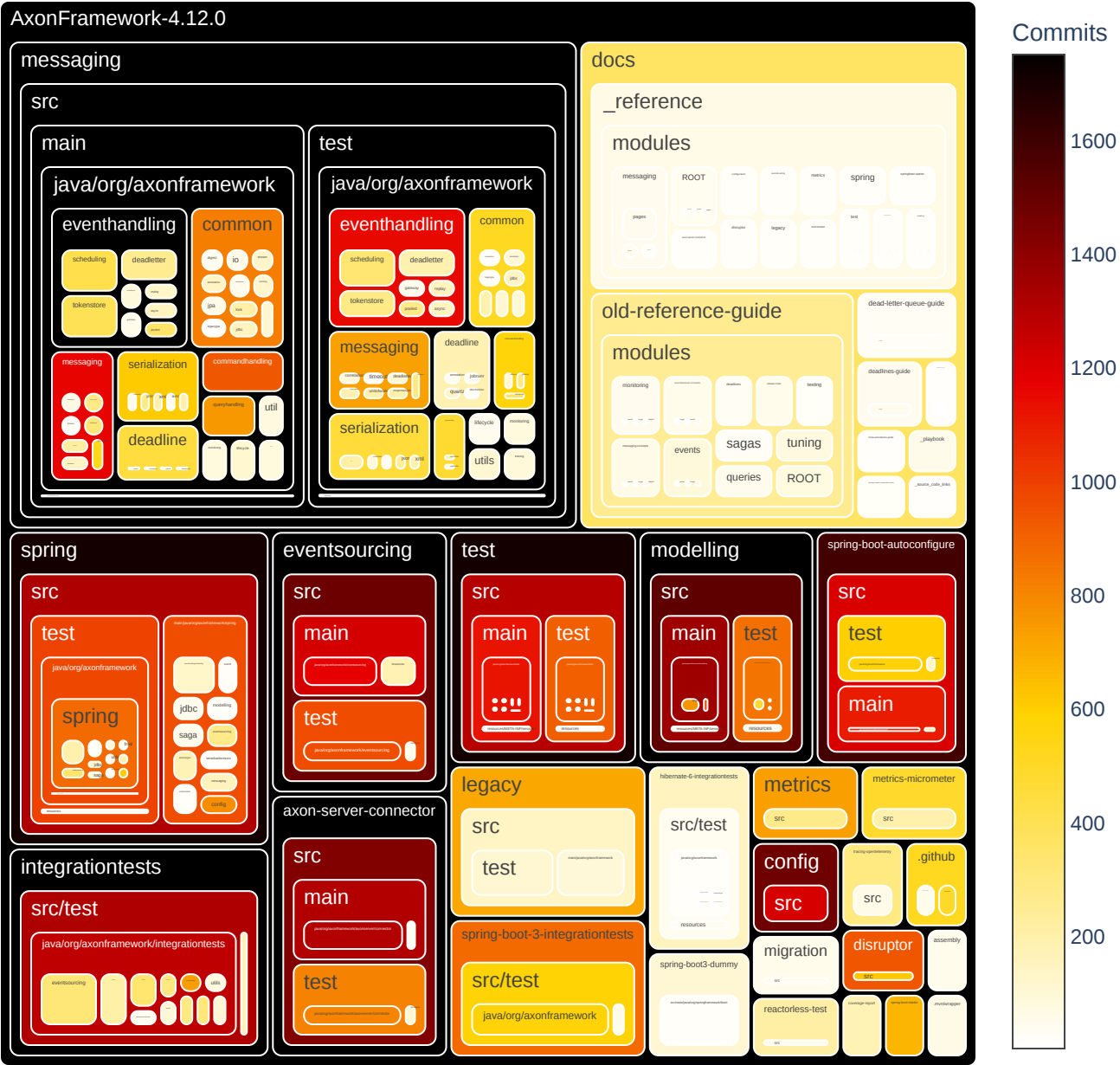
Most frequent file extension per directory

Most frequent file extension per directory



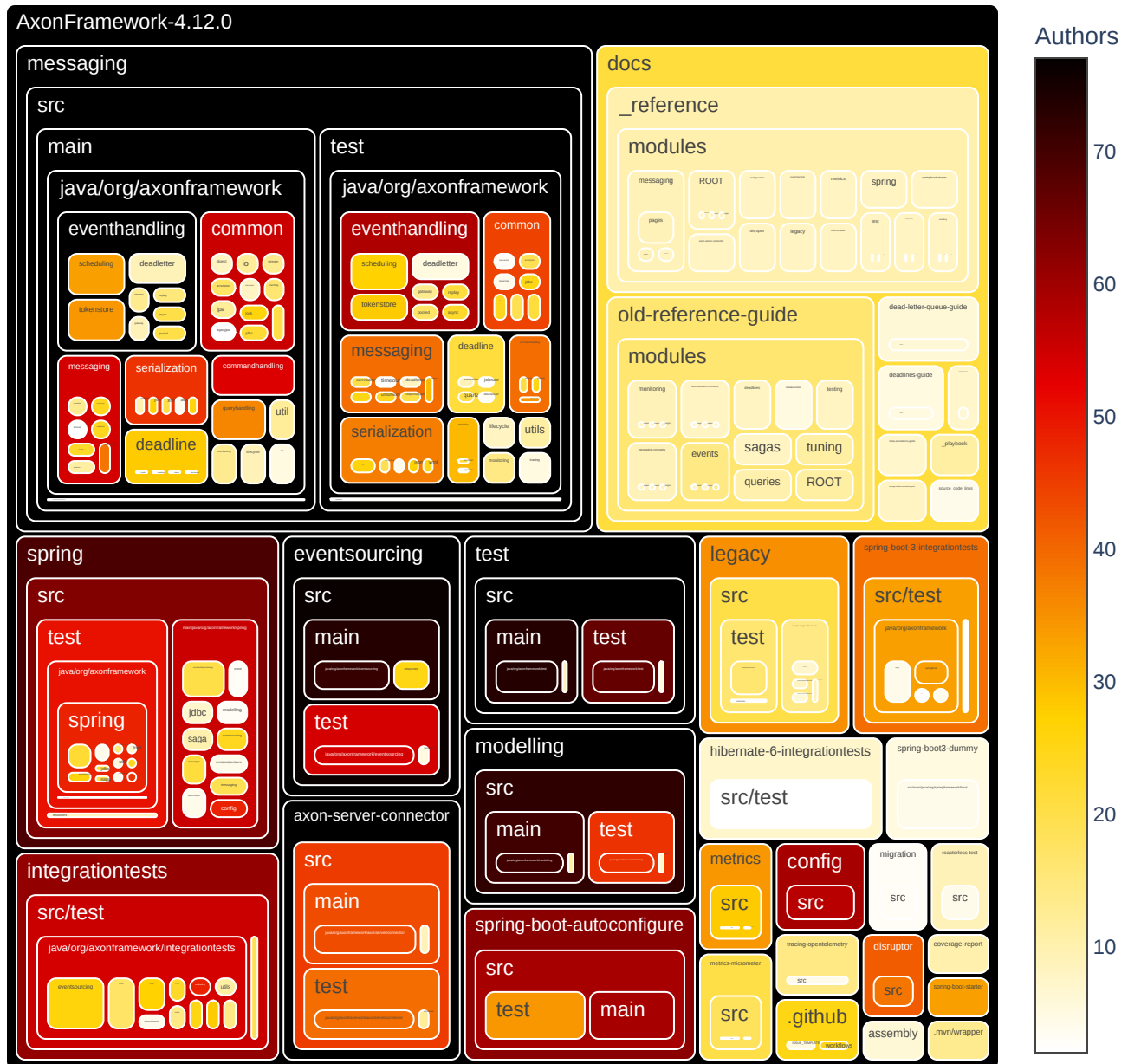
Number of commits per directory

Number of git commits



Number of distinct authors per directory

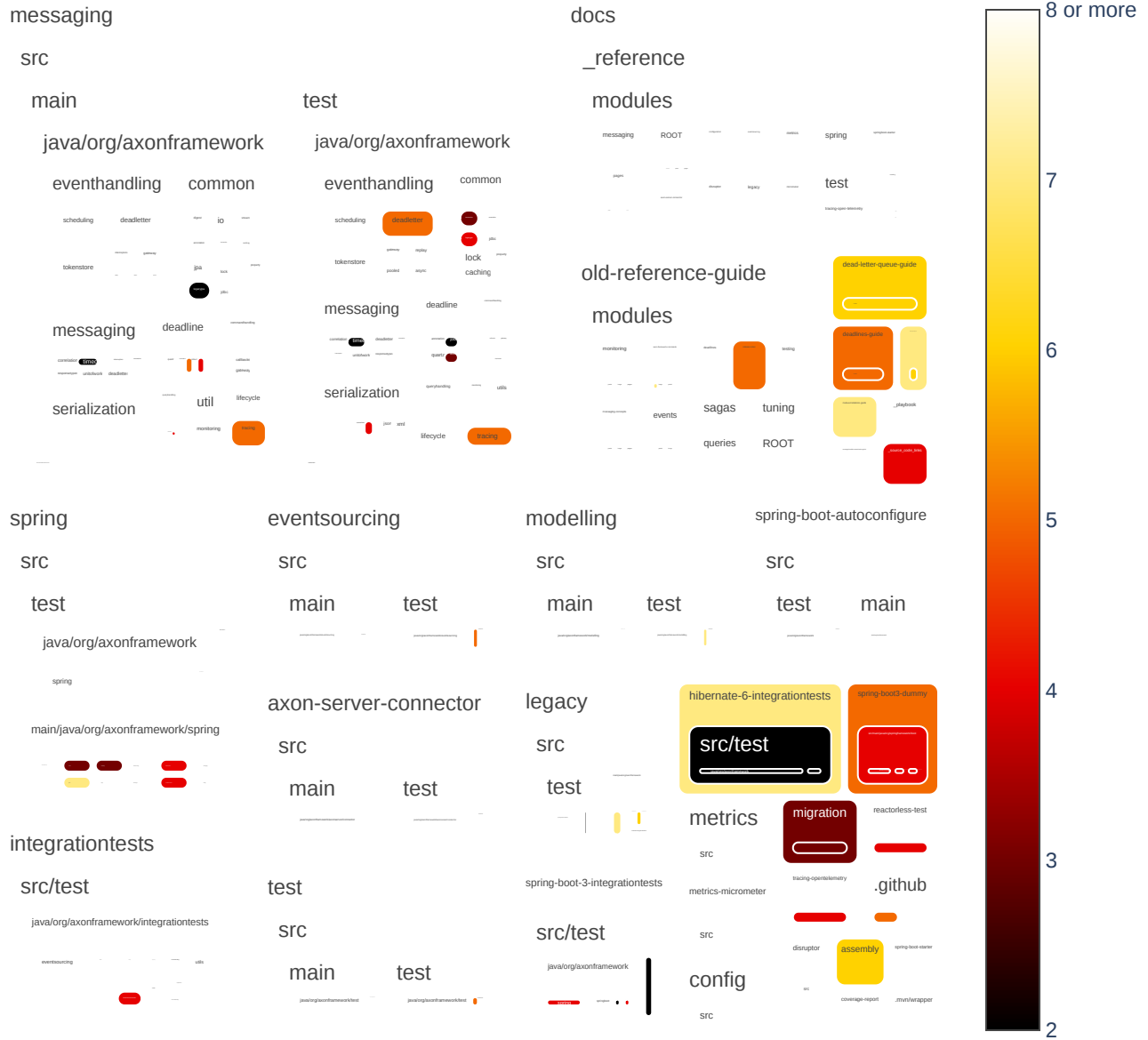
Number of distinct commit authors



Directories with very few different authors

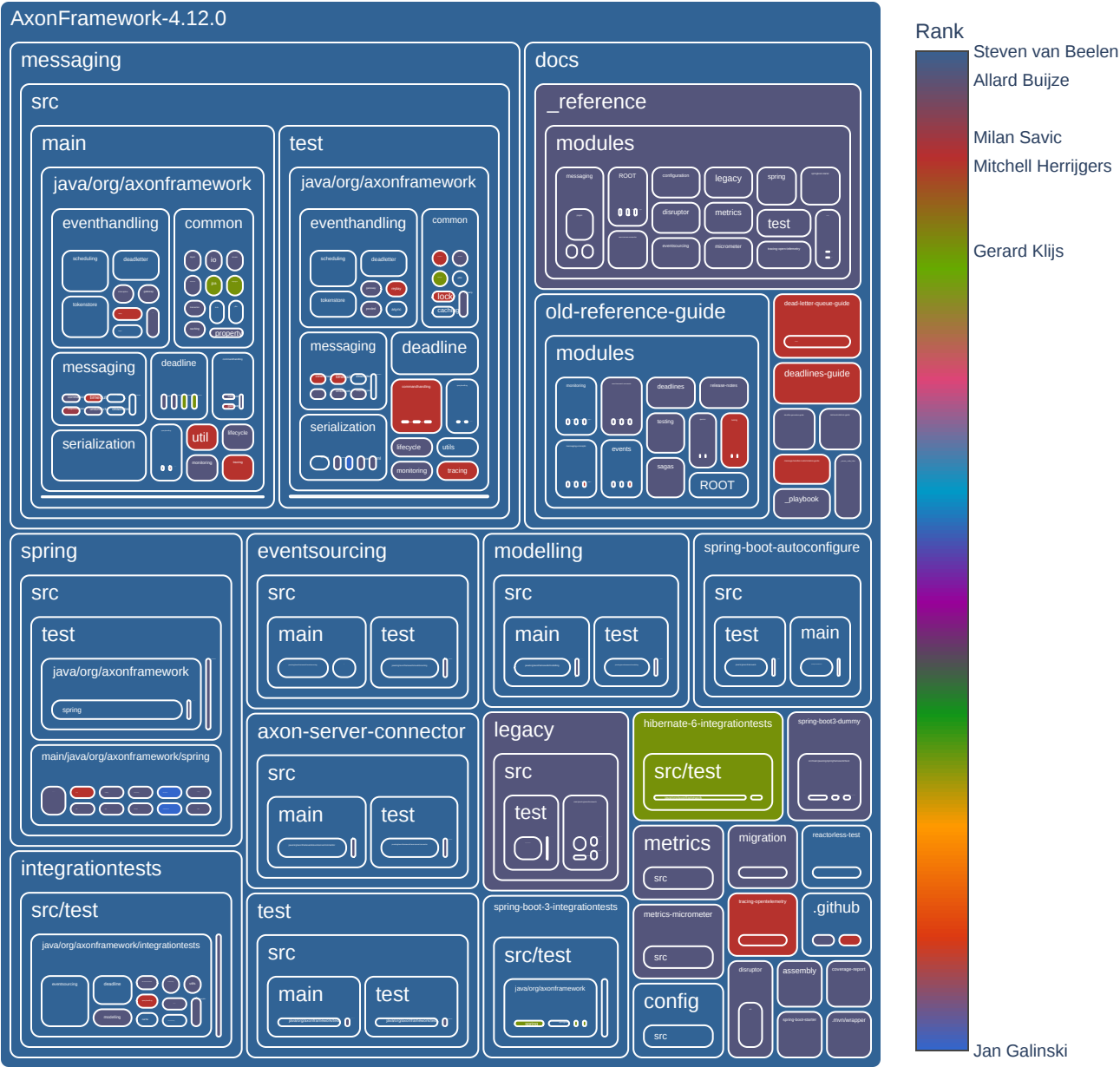
Number of distinct commit authors (red/black = only one or very few authors)

AxonFramework-4.12.0



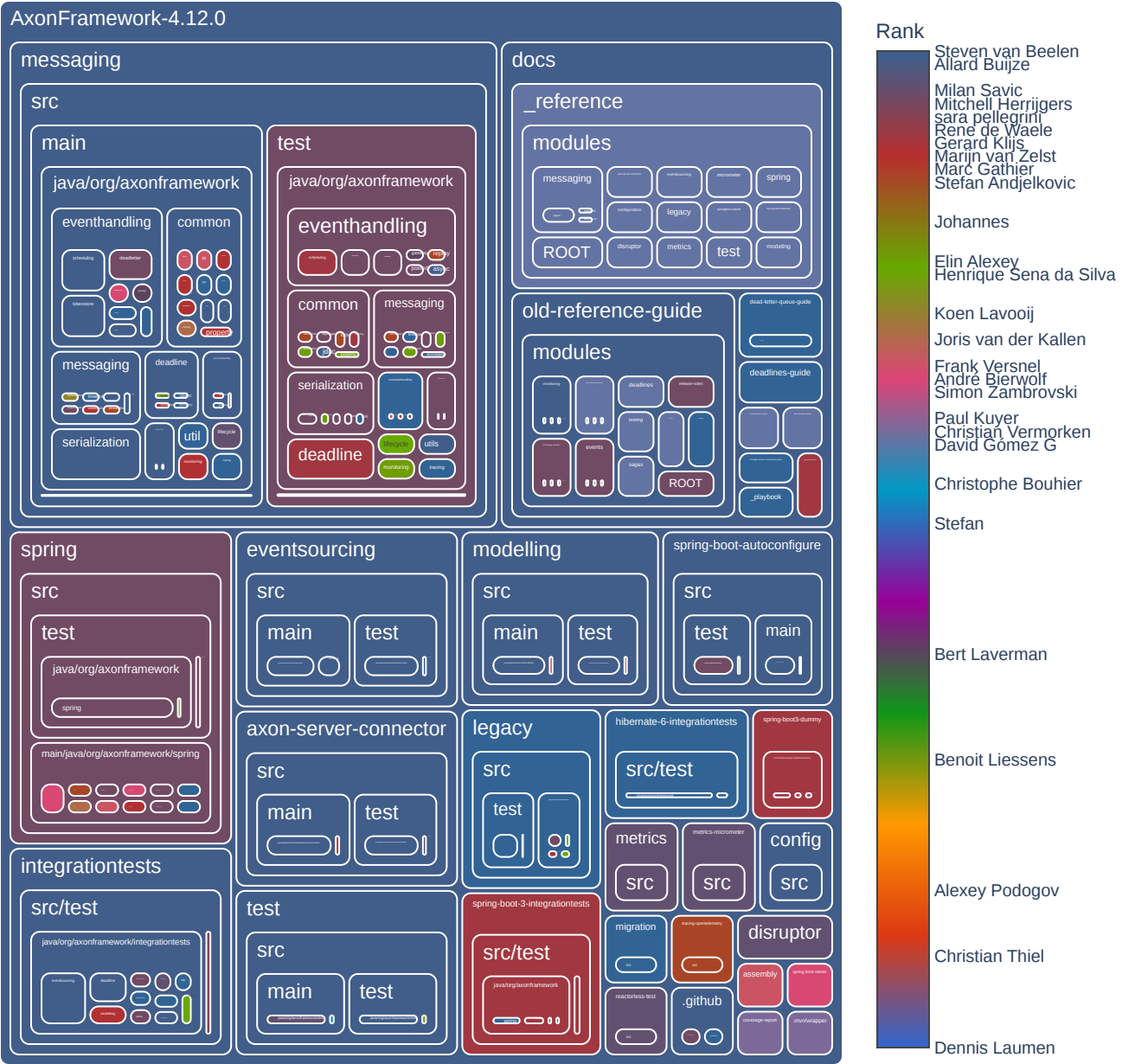
Main author per directory

Main authors with highest number of commits



Second author per directory

Second author with the second highest number of commits



Days since last commit per directory

Days since last commit

AxonFramework-4.12.0

messaging

src

main

java/org/axonframework

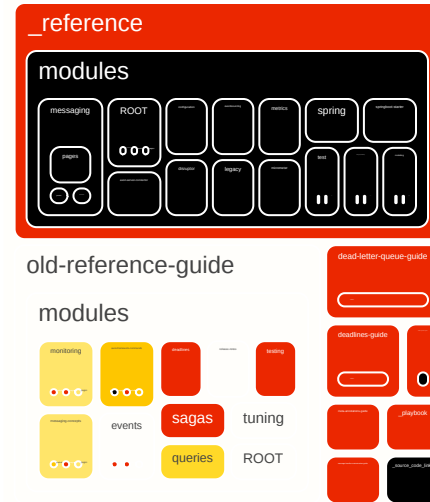


test

java/org/axonframework



docs

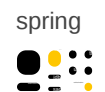


spring

src

test

java/org/axonframework



spring

java/org/axonframework



eventsourcing

src

main

test

java/org/axonframework



test

java/org/axonframework



test

src

main

test

java/org/axonframework



test

java/org/axonframework



modelling

src

main

test

java/org/axonframework



test

java/org/axonframework



spring-boot-autoconfigure

src

test

main

java/org/axonframework



test

java/org/axonframework



integrationtests



axon-server-connector

src

main

test

java/org/axonframework

test

java/org/axonframework

test

java/org/axonframework

test

java/org/axonframework

test

java/org/axonframework

test

java/org/axonframework

test

java/org/axonframework

test

java/org/axonframework

test

legacy

src

test

java/org/axonframework

test

java/org/axonframework

test

java/org/axonframework

test

java/org/axonframework

test

java/org/axonframework

test

java/org/axonframework

test

java/org/axonframework

test

java/org/axonframework

test

java/org/axonframework

test

hibernate-6-integrationtests

src/test

java/org/axonframework

test

java/org/axonframework

test

java/org/axonframework

test

java/org/axonframework

test

java/org/axonframework

test

java/org/axonframework

test

java/org/axonframework

test

java/org/axonframework

test

java/org/axonframework

test

java/org/axonframework

test

metrics

src

config

src

migration

reactorless-test

reactorless-test

reactorless-test

reactorless-test

reactorless-test

reactorless-test

reactorless-test

reactorless-test

reactorless-test

reactorless-test

reactorless-test

reactorless-test

reactorless-test

reactorless-test

reactorless-test

reactorless-test

reactorless-test

metrics-micrometer

src

config

src

disruptor

assembly

assembly

assembly

assembly

assembly

assembly

assembly

assembly

assembly

assembly

assembly

assembly

assembly

assembly

assembly

assembly

assembly

Days

350

300

250

200

150

100

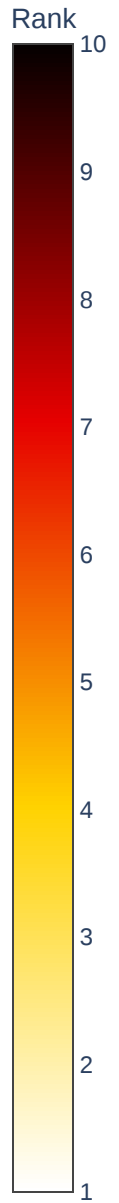
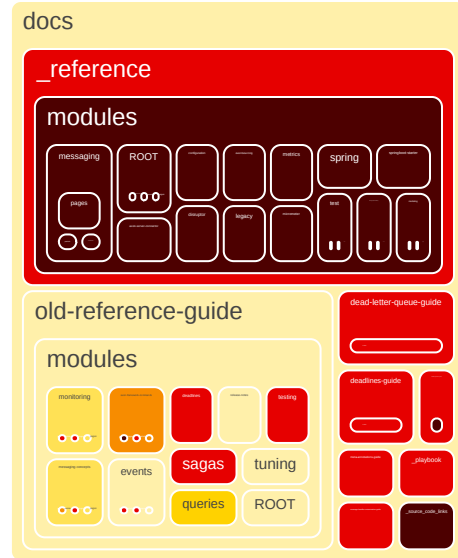
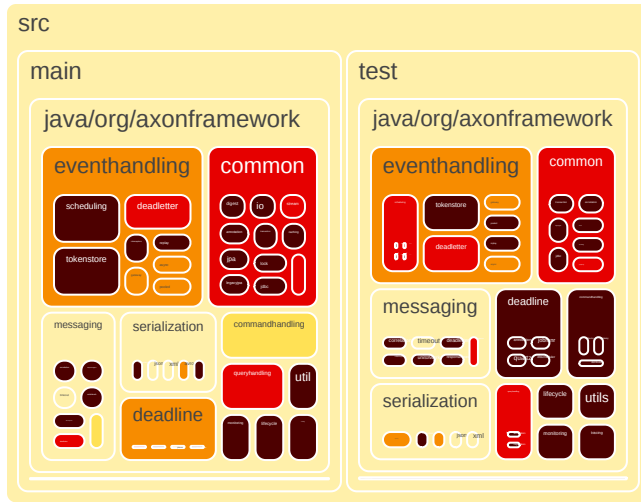
50

Days since last commit per directory (ranked)

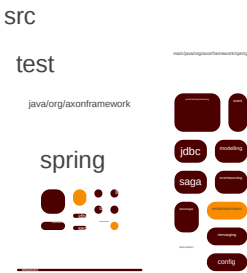
Rank of days since last commit

AxonFramework-4.12.0

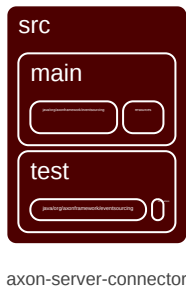
messaging



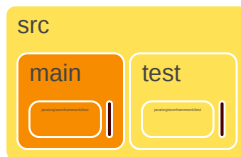
spring



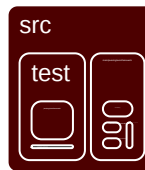
eventsourcing



test



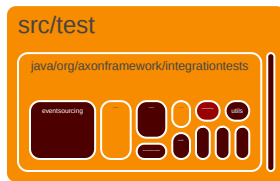
legacy



spring-boot-3-integrationtests



integrationtests

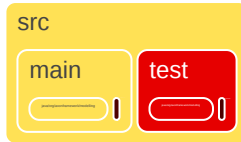


src

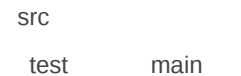
main

test

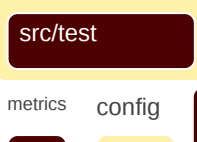
modelling



spring-boot-autoconfigure



hibernate-6-integrationtests



spring-boot3-dummy



metrics

src

src

src

config

src

src

src

migration

src

src

src

reactive-test

src

src

src

coverage-report

spring-boot-starter

assembly

.mvn/wrapper

tracing-opentelemetry

metrics-micrometer

.github

src

src

src

src

src

src

src

src

src

src

src

Days since last file creation

AxonFramework-4.12.0

messaging

src

main

java/org/axonframework



test

java/org/axonframework



docs

_reference

modules



old-reference-guide

modules



spring

src

test



integrationtests

src/test



eventsourcing

src

main



axon-server-connector

src

main

test



test

src

main



legacy

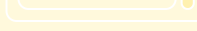
src

test



spring-boot-3-integrationtests

src/test



modelling

src

main



spring-boot-autoconfigure

src

test

main



metrics

src

config

src

migration

reactorless-test



metrics-micrometer

src

config

src

migration

reactorless-test



disruptor

src

config

src

migration

reactorless-test



assembly

src

config

src

migration

reactorless-test



Days

3500

3000

2500

2000

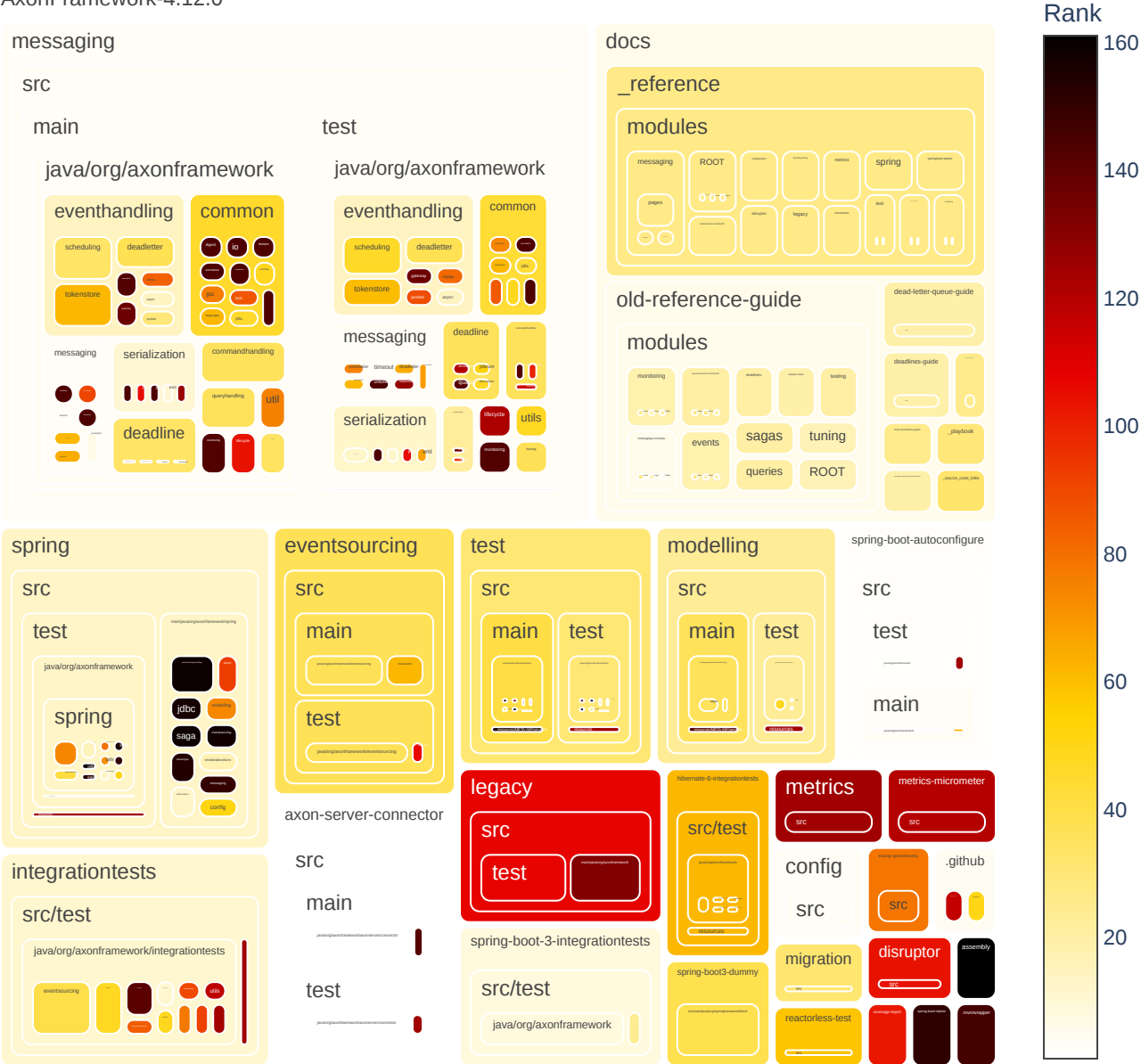
1500

1000

500

Rank of days since last file creation

AxonFramework-4.12.0



Days since last file modification per directory

Days since last file modification

AxonFramework-4.12.0

messaging

src

main

java/org/axonframework

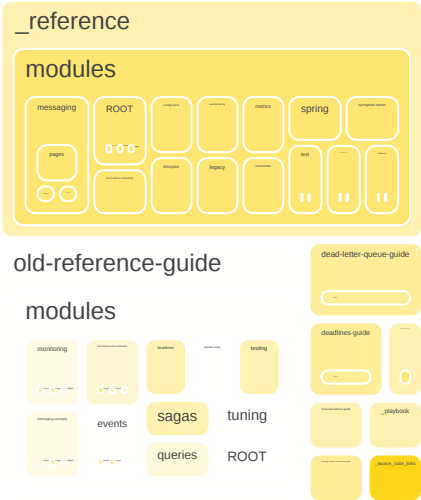


test

java/org/axonframework



docs

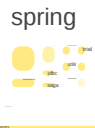


spring

src

test

java/org/axonframework



eventsourcing



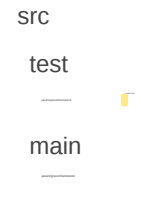
test



modelling



spring-boot-autoconfigure



integrationtests



src

main

test

axon-server-connector

java/org/axonframework/axonserver/connector

legacy



spring-boot-3-integrationtests



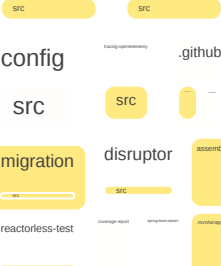
hibernate-6-integrationtests



spring-boot-3-dummy



metrics



Days

2000

1500

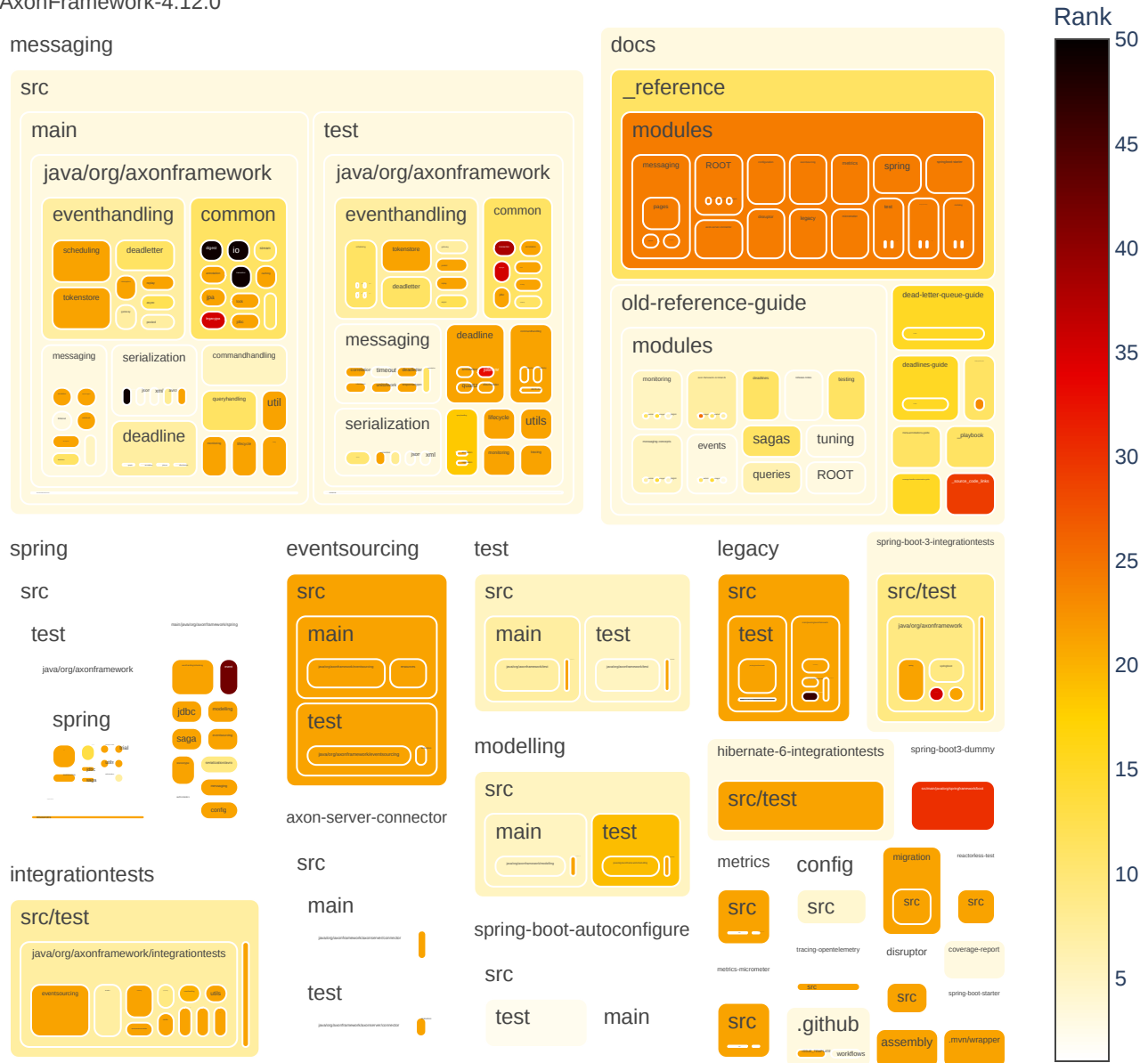
1000

500

Days since last file modification per directory (ranked)

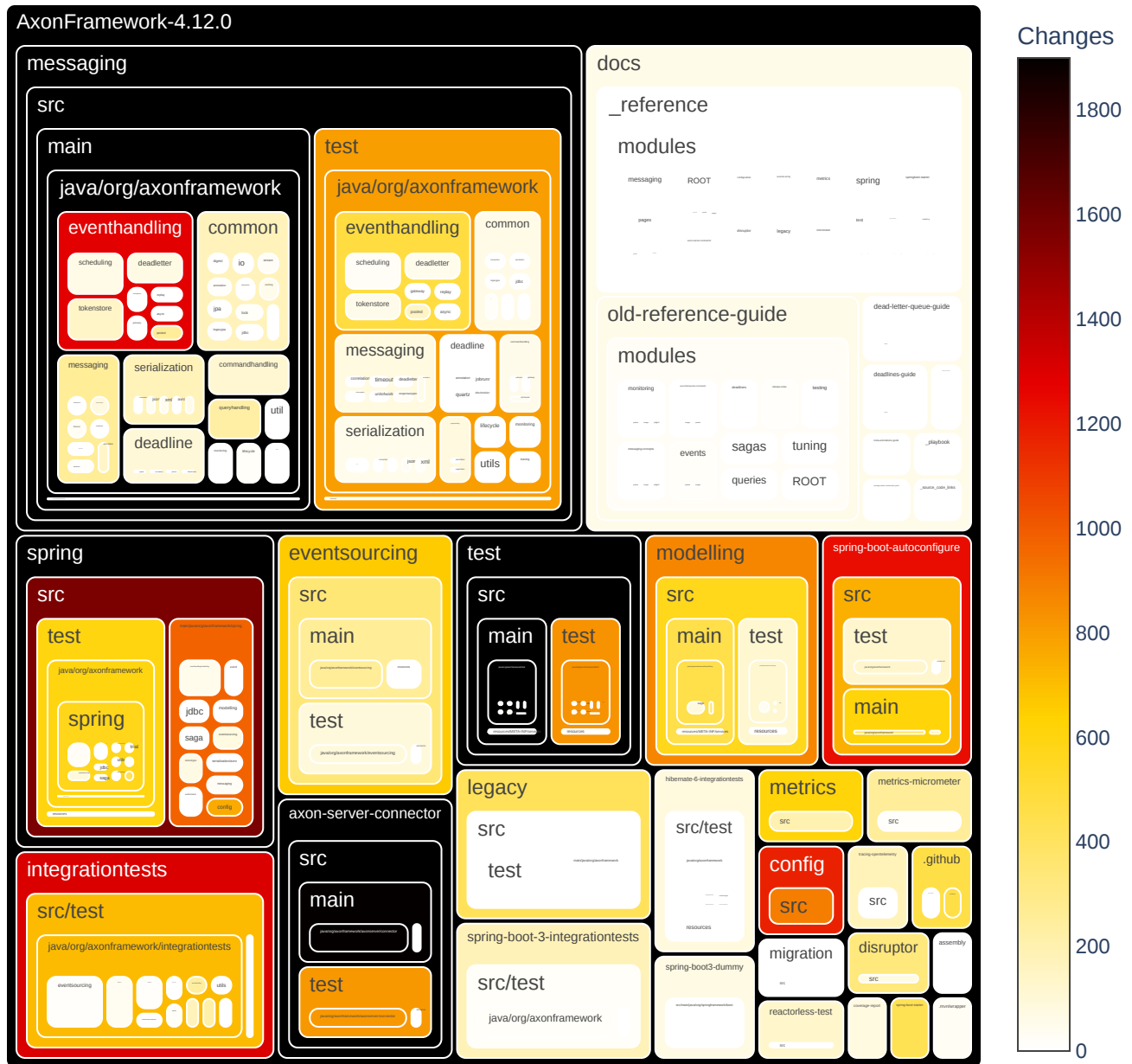
Rank of days since last file modification

AxonFramework-4.12.0



File changed frequently with other files

Pairwise file changes



Filecount per commit

Shows how many commits had changed one file, how many had changed two files, and so on. The chart is limited to 30 lines for improved readability. The data preview also includes overall statistics including the number of commits that are filtered out in the chart.

Preview data

Sum of commits that changed more than 30 files (each) = 971

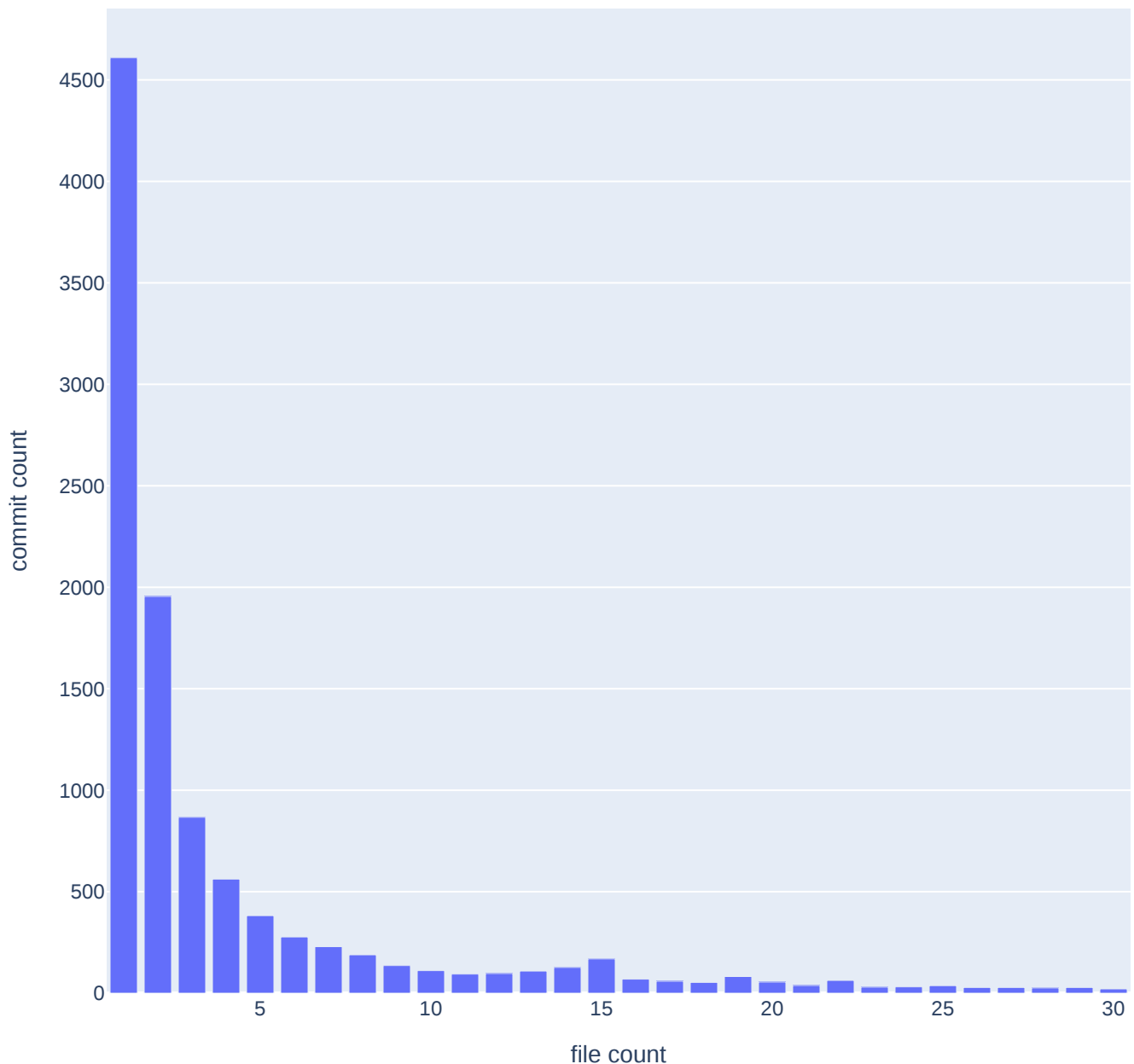
Max changed files with one commit = 2178

	filesPerCommit	commitCount
count	334.000000	334.000000
mean	343.449102	34.526946
std	439.862566	280.454164
min	1.000000	1.000000
25%	84.250000	1.000000
50%	185.000000	2.000000
75%	426.500000	5.000000
max	2178.000000	4609.000000

	filesPerCommit	commitCount
0	1	4609
1	2	1955
2	3	867
3	4	562
4	5	382
5	6	277
6	7	229
7	8	189
8	9	136
9	10	111
10	11	94
11	12	97
12	13	109
13	14	127
14	15	169
15	16	69
16	17	59
17	18	53
18	19	82
19	20	55
20	21	38
21	22	63
22	23	30
23	24	31
24	25	37
25	26	28
26	27	28
27	28	26
28	29	28
29	30	21

Bar chart with the number of files per commit distribution

Changed files per commit



Pairwise Changed Files vs. Dependency Weight

This section explores the correlation between how often pairs of files are changed together (common commit count) and their dependency weight. Note that these results should be interpreted cautiously, as comparing pairwise changes and dependencies is inherently challenging.

Considerations

- **Historical vs. Current State:** Pairwise changes reflect the entire git history, while dependency weight represents the current state of the codebase.

- **Commit Granularity:** Developers may use different commit strategies, such as squashing changes into a single commit or creating fine-grained commits. Ideally, each commit should represent a single semantic change for accurate analysis.
- **Dependency Representation:** Some file types (e.g., Java files with import statements) clearly define dependencies, while others (e.g., shell scripts, XML, YAML) lack explicit dependency relationships.
- **Repository Characteristics:** Repositories with generated code may have many large commits, while stabilized repositories may only update configuration files for dependency changes.

Data Preview

```
Received notification from DBMS server: {severity: WARNING} {code: Neo.ClientNotification.Statement.UnknownPropertyKeyWarning} {category: UNRECOGNIZED} {title: The provided property key is not in the database} {description: One of the property names in your query is not available in the database, make sure you didn't misspell it or that the label is available when you run this statement in your application (the missing property name is: cardinality)} {position: line: 9, column: 48, offset: 576} for query:
"// List pair of files that were changed together and that have a declared dependency between each other.\n\nMATCH (firstCodeFile:File)-[dependency:DEPENDS_ON]->(secondCodeFile:File)\nMATCH (firstCodeFile)-[pairwiseChange:CHANGED_TOGETHER_WITH]-(secondCodeFile)\n//De-duplicating the pairs of files isn't necessary, because the dependency relation is directed.\n//WHERE elementId(firstCodeFile) < elementId(secondCodeFile)\nWITH firstCodeFile.fileName AS firstFileName\n,secondCodeFile.fileName AS secondFileName\n,coalesce(dependency.weight, dependency.cardinality) AS dependencyWeight\n,pairwiseChange.commitCount AS commitCount\n,dependency.fileDistanceAsFewestChangeDirectoryCommands AS fileDistanceAsFewestChangeDirectoryCommands\nRETURN dependencyWeight\n,commitCount\n,fileDistanceAsFewestChangeDirectoryCommands\n// ,count(*) AS occurrences\n// ,collect(firstFileName + ' -> ' + secondFileName)[0..3] AS examples\nORDER BY dependencyWeight, commitCount\n\n// MATCH (firstCodeFile:File)-[dependency:DEPENDS_ON]->(secondCodeFile:File)\n// MATCH (firstCodeFile)-[pairwiseChange:CHANGED_TOGETHER_WITH]-(secondCodeFile)\n// WHERE elementId(firstCodeFile) < elementId(secondCodeFile)\n// RETURN firstCodeFile.fileName AS firstFileName\n// ,secondCodeFile.fileName AS secondFileName\n// ,dependency.weight AS dependencyWeight\n// ,pairwiseChange.commitCount AS commitCount\n// ORDER BY dependencyWeight, commitCount\n\n// MATCH (g1:!Git&File)-[relation:CHANGED_TOGETHER_WITH|DEPENDS_ON]-(g2:!Git&File)\n// WITH count(DISTINCT relation) AS relatedFilesCount\n// ,collect(DISTINCT relation) AS relations\n// UNWIND relations AS relation\n// WITH relatedFilesCount\n// ,coalesce(relation.commitCount, 0) AS commitCount\n// ,coalesce(relation.weight, 0) AS dependencyWeight\n// ,coalesce(relation.fileDistanceAsFewestChangeDirectoryCommands, 0) AS fileDistanceAsFewestChangeDirectoryCommands\n// RETURN dependencyWeight\n// ,commitCount\n// ,fileDistanceAsFewestChangeDirectoryCommands\n// ORDER BY dependencyWeight, commitCount"
```

	dependencyWeight	commitCount	fileDistanceAsFewestChangeDirectoryCommands
0	1	12	0
1	1	12	0
2	1	12	4
3	1	12	0
4	1	12	0
5	1	12	4
6	1	12	0
7	1	12	2
8	1	12	0
9	1	12	4
10	1	13	0
11	1	13	0
12	1	13	4
13	1	13	4
14	1	14	4
15	1	14	2
16	1	15	0
17	1	15	2
18	1	15	0
19	1	16	0

Data Statistics

'Pairwise changed git files compared to dependency weights - Overall statistics'

	dependencyWeight	commitCount	fileDistanceAsFewestChangeDirectoryCommands
count	167.000000	167.000000	167.000000
mean	9.694611	25.251497	1.365269
std	19.250321	16.694894	1.569127
min	1.000000	12.000000	0.000000
25%	1.500000	14.000000	0.000000
50%	4.000000	19.000000	0.000000
75%	10.000000	28.000000	2.000000
max	154.000000	95.000000	4.000000

'Pairwise changed git files compared to dependency weights - Pearson Correlation'

	dependencyWeight	commitCount	fileDistanceAsFewestChangeDirectoryCommands
dependencyWeight	1.000000	0.204498	-0.137682
commitCount	0.204498	1.000000	-0.138284
fileDistanceAsFewestChangeDirectoryCommands	-0.137682	-0.138284	1.000000

'Pairwise changed git files compared to dependency weights - Spearman Correlation'

	dependencyWeight	commitCount	fileDistanceAsFewestChangeDirectoryCommands
dependencyWeight	1.000000	0.211049	-0.083460
commitCount	0.211049	1.000000	-0.124036
fileDistanceAsFewestChangeDirectoryCommands	-0.083460	-0.124036	1.000000

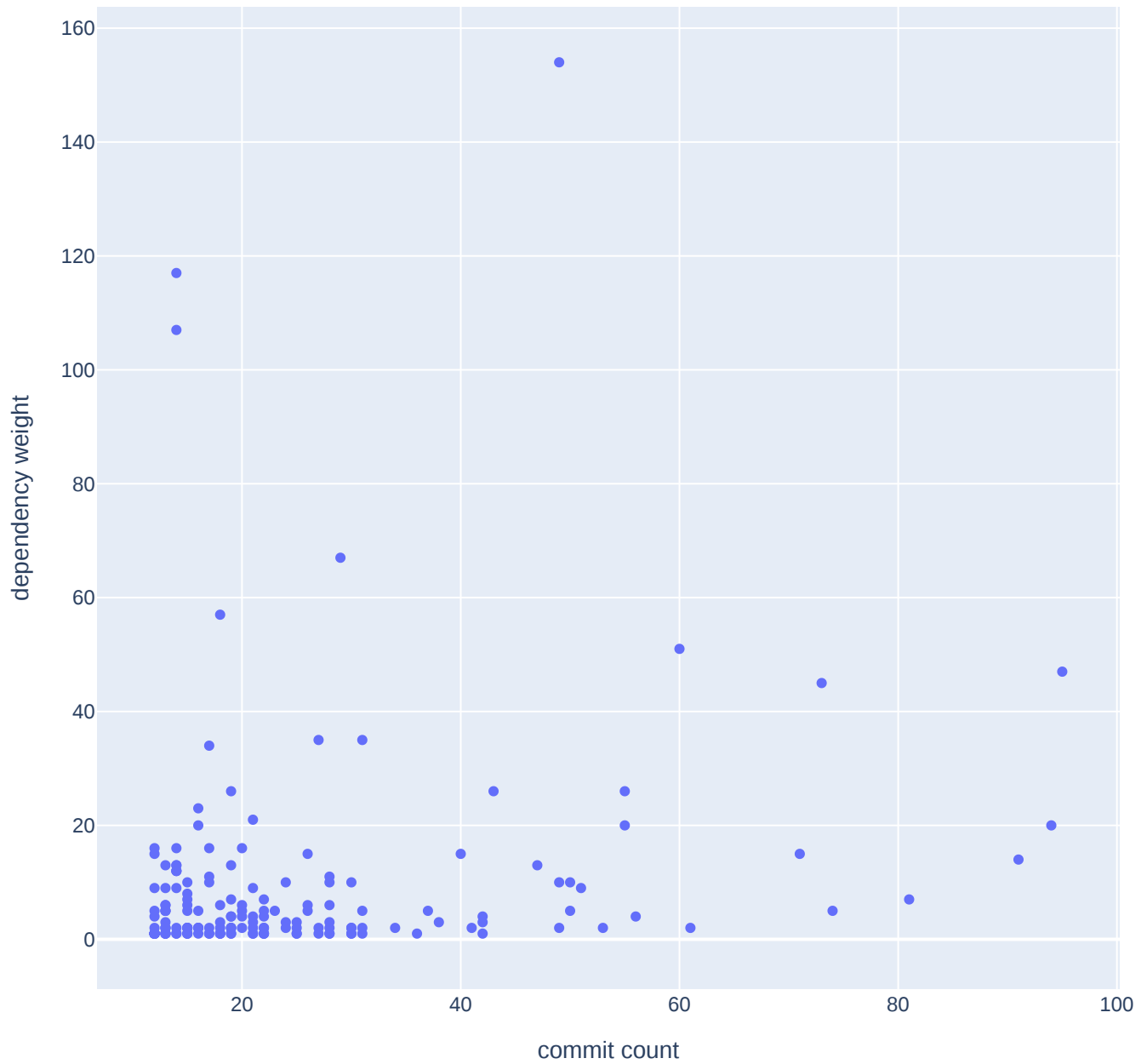
'Pearson Correlation with p-value for commitCount and dependencyWeight'

PearsonRResult(statistic=0.20449754623098543, pvalue=0.008027370650833442)

'Spearman Correlation with p-value for commitCount and dependencyWeight'

SignificanceResult(statistic=0.2110488339508042, pvalue=0.006185645631183674)

Pairwise changed files: Number of changes (commitCount) vs. dependency weight



WordCloud of git authors

	word	frequency
0	Steven van Beelen	3525
1	Allard Buijze	2724
2	smcvb	822
3	Milan Savic	449
4	Mitchell Herrijgers	327
5	Gerard Klijs	246
6	Marc Gathier	231
7	sara pellegrini	163
8	Rene de Waele	121
9	Jetro Coenradie	117

Wordcloud of git authors

